

Valentine Agam

Miri, Sarawak | 014-6521429 | valentineagam6@gmail.com

LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Backend and AI-focused software engineer with hands-on experience developing scalable web applications, automation systems, and computer vision solutions using Python, FastAPI, Next.js, TensorFlow, and Supabase. Experienced in building AI-powered recruitment systems, workflow automation, and object detection models through freelance and contract-based projects. Strong foundation in backend architecture, API development, and machine learning deployment.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL, JavaScript
- **Frontend:** React.js, Next.js, Flutter
- **Backend:** FastAPI, REST API Development, Supabase
- **AI/ML:** TensorFlow, YOLO, OpenCV, Scikit-learn
- **Database & Cloud:** PostgreSQL, Firebase, DigitalOcean
- **Tools & Technologies:** Git, Power BI, QGIS, Global Mapper, n8n Automation
- **Concepts:** RBAC Security, Microservices, OCR, Computer Vision

WORK EXPERIENCE

Freelance Full-Stack Developer

Miri, Sarawak | Mar 2026 – Present

- Developed responsive full-stack web applications for client-based projects using modern frontend and backend technologies.
- Designed mobile-friendly and desktop-optimized user interfaces with strong focus on usability and performance.
- Developed scalable backend APIs, authentication systems, and database integrations using Python, FastAPI, and Supabase.
- Collaborated directly with clients to gather requirements and maintain deployed applications.

Junior Backend Engineer - Zenara Jaya

Miri, Sarawak | Dec 2025 – Mar 2026

- Developed an AI-powered Applicant Tracking System (ATS) using FastAPI and Next.js to automate recruitment workflows.
- Implemented resume parsing with Claude API, OCR fallback, and PDF extraction using PyMuPDF and pdfplumber.
- Built RBAC authentication and permission systems for Admin, Recruiter, and Client Manager roles.
- Automated recruitment operations using n8n workflows, reducing manual processing time by 60%.
- Deployed scalable backend services on DigitalOcean with Supabase PostgreSQL integration and cloud storage.

Geographic Information Systems (GIS) Intern - Sarawak Oil Palm

Miri, Sarawak | Jul 2024 – Oct 2024

- Conducted GIS digitization and palm census analysis using QGIS for operational planning and land management.
- Developed Python automation scripts for FIT-to-GPX conversion to improve field data processing efficiency.
- Built TensorFlow and OpenCV-based object detection solutions for automated soil hole identification.
- Produced 3D spatial visualization videos using Global Mapper to support data presentation and analysis.

PROJECTS

Oil Palm Object Detection (TensorFlow, YOLO, OpenCV)

- Trained custom YOLO object detection models achieving 92% accuracy for oil palm tree and soil hole detection.
- Applied image augmentation and dataset optimization techniques to improve detection robustness.
- Developed automated detection workflows using Python and OpenCV.

AI Recruitment System (FastAPI, Next.js, Supabase)

- Developed an AI-powered recruitment platform with resume parsing, OCR fallback, and role-based authentication.
- Integrated Claude API for automated candidate data extraction and workflow automation.
- Built scalable backend APIs and cloud-based storage infrastructure.

AR Navigation Application (Flutter, Unity, WebAR)

- Built a mobile AR navigation application for campus exploration using Flutter SDK and Unity integration.
- Developed interactive WebAR features to improve navigation and user engagement.

EDUCATION

University Malaysia Sarawak (UNIMAS) | Graduated Nov 2025

Bachelor of Science in Cognitive Science (Minor in Computer Science)

- Dean's List (Year 3 Semester 2)
- Relevant Coursework: Artificial Intelligence, Data Analysis, Software Development, Web Development

AWARDS & CERTIFICATIONS

- **CompTIA Security+** - CompTIA
- **Python for Data Science** - Cognitive Class

LANGUAGES

English – Fluent in verbal and writing

Bahasa Melayu – Fluent in verbal and writing

REFERENCES

Zenara Jaya - Mohammad Aiman Bin Alwi | Chief Technology Officer | Phone Number: 017 806 5862

Sarawak Oil Palms (SOPB) - Jane Mujan | Assistant Manager GIS | Phone Number: 017 805 0983